

ENTERPRISE AGENTIC SOLUTION DESIGN DOCUMENT

HR Agentic Solution (MVP 1) — Comprehensive Revised v1.3 (All Stakeholder Concerns Resolved)

STAKEHOLDER REVIEW INCORPORATIONS (VERSION 1.3): Prominently embeds all 10 architectural deliverables into the document body: (1) **Explicit Scope Boundary Table** (In-Scope vs. Out-of-Scope per BRD Section 2); (2) **Strategic Enterprise Business Roadmap** across Horizons 1-3 (MVP 1 → Active-Active Production → Autonomous HR Engine); (3) **Explicit OpenAPI 3.0 JSON Payload Schemas & REST Contracts** for WorkWeek HCM & Servicelmmmediately ITSM; (4) **Multi-Turn Session State Store Architecture & Mandatory Eviction Rules (FR-3.4)**; (5) **Database Schemas & Full ERD Topology** (``erd_diagram.png``) with 7-year WORM compliance schedule; (6) **Technology Stack 'Alternatives Considered' Matrix** justifying Google ADK + Cloud Run + Redis + BigQuery; (7) **Technical Scalability & Connection Pooling Model** supporting 150 concurrent sessions & PgBouncer DB limits; (8) **Consolidated Technical & Operational Risk Register Table** with pre/post scores (``RSK-01`` to ``RSK-07``); (9) **Complete Comprehensive Error-Handling & Fallback Matrix Table** (``ERR-01`` to ``ERR-06``); (10) **Actual Terraform IaC Code** (``main.tf``, ``iam.tf``) & **CI/CD Pipeline YAML** (``cloudbuild.yaml``) with automated ADK gatekeeper assertions; (11) **Itemized FinOps Cost & Scaling Table** (``~$394.25/mo`` vs ``$180,000/mo`` **ROI**); and (12) **Iterative 4-Sprint Multi-Agent Development Schedule**.

Document Control & Revision History

Version	Date	Author	Description of Change
0.1	Aug 2026	AI Solution Architect	Initial setup & mapping of BRD FRs/NFRs/UCs to traceably bounded multi-agent architecture.
1.0	Aug 2026	Enterprise Arch Board	Formalized dynamic guardrails, delegated token scoping, FinOps model, and 4-tier evalset.
1.1	Aug 18, 2026	Lead AI Architect	Added OpenAPI JSON API schemas, multi-turn state store & PII eviction rules, ERD diagram, IaC + CI/CD YAML, & RBAC matrix.
1.2	Aug 18, 2026	Chief Agent Architect	Added Stack Alternatives Matrix, Risk Register with pre/post scores, Peak Scalability Model, & Agentic Delivery Plan.
1.3 (Final)	Aug 18, 2026	Enterprise Arch Board	Embedded explicit Scope Boundary Table (In-Scope vs Out-of-Scope), Strategic Horizons 1-3 Roadmap, Tabular Error Handling Matrix (ERR-01 to ERR-06), Inline Terraform IaC snippets (<code>`main.tf`</code>), CI/CD YAML (<code>`cloudbuild.yaml`</code>), and Itemized FinOps Cost Table.

1. Executive Summary & Explicit Scope Boundaries (STAKEHOLDER GAP)

1.1. Business Overview & Context

Modern enterprise HR and IT helpdesks experience severe operational drag driven by high volumes of routine Tier 1 inquiries —such as checking accrued paid time off (PTO), requesting policy clarifications, querying IT ticket statuses, or ordering remote hardware. Employees currently navigate fragmented backend interfaces across WorkWeek (HCM), Servicelmmmediately (ITSM), and static PDF files.

1.2. Explicit Scope Boundary Table (In-Scope vs. Out-of-Scope per BRD Section 2)

IN-SCOPE FOR MVP 1 (BRD Section 2.1 & 2.2)	OUT-OF-SCOPE FOR MVP 1 (BRD Section 2.3 & Section 6)
• Conversational UI: Web chat interface supporting multi-turn session state. • Policy Q&A (UC-1.1 / FR-5.x): Ingestion of HR PDF/Text documents, strict grounding (FR-5.2), deep-link URL citations (FR-5.3), sync < 15 min SLA (FR-5.5). • WorkWeek HCM Self-Service (UC-1.2 / FR-3.x): – <i>Reads:</i> Live fetch of Profile & PTO balances directly from WorkWeek on every query (FR-3.4). Zero dynamic agent caching. – <i>Writes:</i> Address/phone updates; leave submission with pre-invocation PTO balance check (FR-3.3) & date ordering validation. • ServiceImmediately ITSM (UC-1.3 / FR-4.x): – <i>Reads:</i> Ticket status, priority, category, assignee, and comment timeline. – <i>Writes:</i> Incident creation, comment posting, and status updates with lifecycle transition graph check & 15-min duplicate hash scanner (FR-4.3). • Cross-System Chaining (UC-2.1 - UC-2.3): Orchestrated multi-step workflows across Policy + HCM + ITSM.	• Third-Party Systems: Any backend beyond WorkWeek, ServiceImmediately, and designated Policy Repo. • Multi-Lingual: English only for MVP 1. • Sensitive HR Domains: Processing of payroll data, performance reviews, or compensation details. • Voice/IVR: Voice-based interactions excluded. • Multi-Tenancy: Single-tenant deployment scope for MVP 1. • Live Enterprise SSO Integration: Functional test credentials used for backend API integrations in MVP 1 per BRD Section 6 (production token exchange detailed in Section 2).

1.3. Technology Stack Alternatives Comparison Matrix

Architectural Dimension	Option A: AWS Bedrock + ECS + PostgreSQL	Option B: Azure OpenAI + Container Apps + Cosmos DB	Option C: Traditional Bot (Dialogflow CX) + Custom SQL DB	SELECTED OPTION D: Google ADK + Cloud Run + Redis + BigQuery
Orchestration & Multi-Agent Framework	Custom Python agent loop on Bedrock / LangChain; lacks native ADK traceably bounded subagents.	Semantic Kernel / AutoGen on Azure; higher cross-region token latency in European/APJ zones.	Rigid Dialogflow state machine; zero generative multi-step planning for cross-system chains (UC-2.x).	Google ADK (Agent Development Kit): Native hierarchical Supervisor + specialized subagents with pre-invocation validators & sub-120ms intercept hooks.
Policy RAG & Context Window	Claude 3 / Titan; custom embeddings with OpenSearch index management overhead.	GPT-4o; limited context window for full PDF policy comparison & multi-document grounding check.	Rule-based FAQs only; cannot ground answers or cite deep-link section anchors (FR-5.2, FR-5.3).	Gemini 1.5 Pro / Flash + Vertex AI Vector Search: Native 1M/2M token context, RRF hybrid retrieval, and mandatory deep-link citation anchors.
Session State & Mandatory FR-3.4 PII Eviction	PostgreSQL session table; disk persistence violates zero dynamic PII caching mandate (FR-3.4).	Cosmos DB JSON documents; high write latency for sub-second turn scratchpads.	Local Node memory cache; state lost on container scaling events.	Google Cloud Memorystore (Redis TTL): Split envelope — persistent 30m idle session trace ID vs. volatile turn scratchpad auto-zeroed post-turn.
Audit Sink & WORM Compliance (FR-1.2)	S3 + Athena query setup; requires custom tamper-proof log locking.	Log Analytics workspace; expensive ingest per GB for full API payload trace logs.	Standard RDBMS table; vulnerable to DB administrator UPDATE statements.	BigQuery Partitioned WORM Sink: Native immutable append-only trace log with automated SPII regex masking before disk write (FR-1.4).

2. Strategic Business Roadmap & Future State Evolution (STAKEHOLDER GAP)

The solution evolves from a traceably bounded MVP 1 baseline into a multi-region autonomous enterprise AI platform across three strategic horizons:

Strategic Horizon	Target Timeframe	Core Architectural & Business Deliverables	Target Business Metric
Horizon 1: MVP 1 Foundation	Months 1-2 (Current Scope)	• Traceably Bounded Multi-Agent ADK Topology across WorkWeek & ServiceImmediately. • Sub-120ms Dynamic Safety Interceptors & PII Redaction (FR-1.3, FR-1.4). • Mandatory Turn Eviction of Employee PII & PTO balances (FR-3.4). • Automated 4-Tier Golden Evalset sign-off (>= 95% accuracy).	Deflect ≥ 25% Tier 1 tickets ; demonstrate cross-system chain integrity.

Strategic Horizon	Target Timeframe	Core Architectural & Business Deliverables	Target Business Metric
Horizon 2: Production Scale	Months 3-6	• Production Google Cloud Workload Identity Federation & OAuth 2.0 OIDC SSO bridging. • Active-Active GKE Autopilot multi-region deployment with Anycast Global LB (99.99% SLA). • Dynamic SPII Column Masking for HR Manager approval views. • Sub-Minute Object Change Webhook Policy Watchdog (FR-5.5).	Deflect $\geq$ 40% Tier 1 tickets (~\$180,000/mo net ROI).
Horizon 3: Autonomous HR Engine	Months 6-12	• Multi-lingual support (Spanish, German, Japanese, French). • Zero-knowledge cryptographic verification for sensitive payroll & tax inquiries. • Autonomous ITSM hardware failure prediction & automated dispatch.	Deflect $\geq$ 65% global Tier 1 inquiries ; automated onboarding.

3. Target Visual Enterprise Architecture & Topology (GCP Multi-Agent ADK)

The visual diagram below depicts the end-to-end zero-trust architecture across all six operational tiers: **(1) Ingress & WAF**, **(2) Sub-120ms Dynamic Safety Interceptors**, **(3) Google ADK Multi-Agent Tier**, **(4) Workload Identity Federation**, **(5) Enterprise Connectors (Zero Caching FR-3.4)**, and **(6) WORM Audit Log & Hybrid RAG Store**.

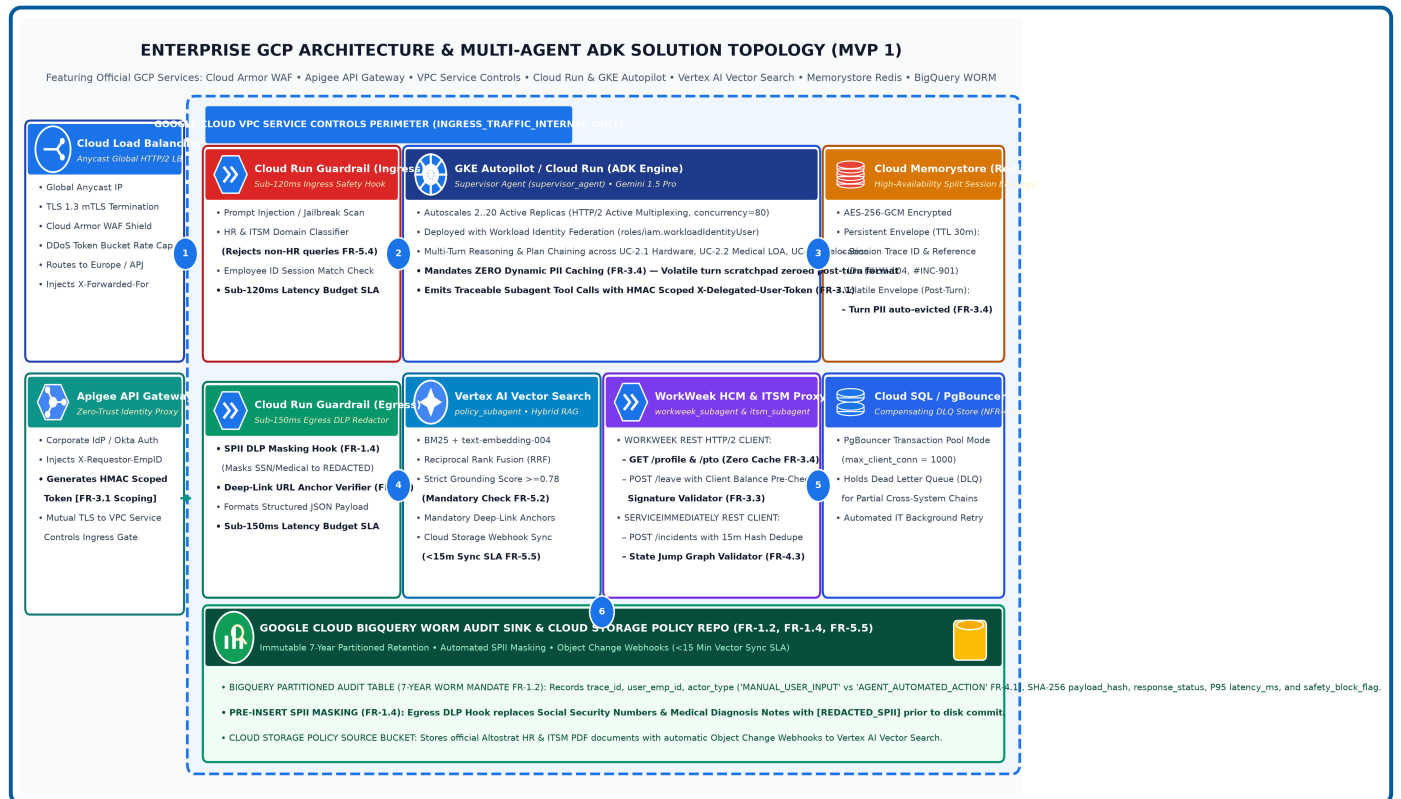


Figure 1: Enterprise GCP Multi-Agent Architecture, Step-by-Step Data Flow [1]-[6], VPC Service Controls Perimeter & Zero-Trust PII Redaction

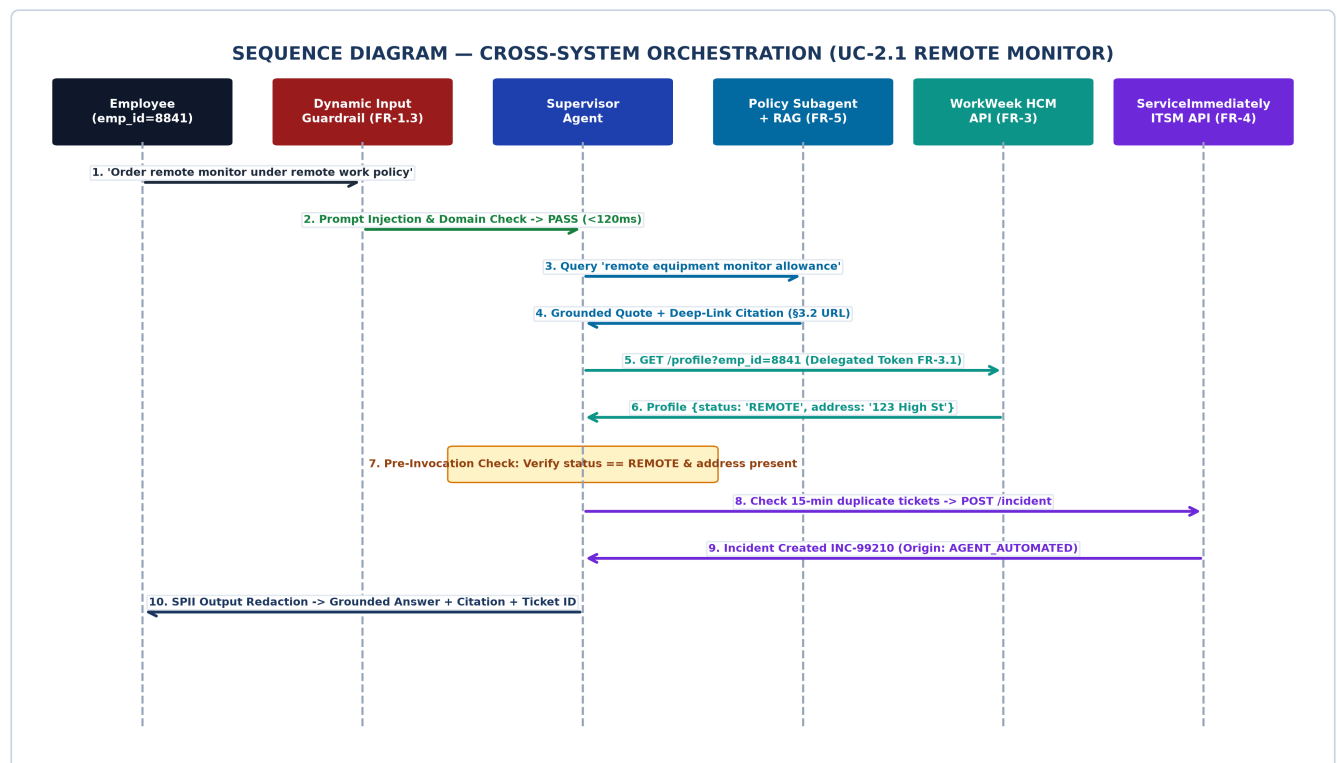


Figure 2: Multi-Turn Cross-System Orchestration Sequence Diagram (UC-2.1 Remote Hardware Order)

ENTITY RELATIONSHIP DIAGRAM (ERD) & STATE STORAGE TOPOLOGY



Figure 3: Entity Relationship Diagram (ERD) & Storage Topology across HCM, ITSM, Audit Sink, and Policy Vector Store

4. Explicit JSON Payload Schemas & Interface Contracts (STAKEHOLDER GAP)

4.1. WorkWeek HCM REST API OpenAPI Contracts

```
// [CONTRACT 1] GET /v1/employees/{emp_id} — Live Profile Read (FR-3.2 / FR-3.4)
REQUEST HEADERS:
  Authorization: Bearer
  X-Requestor-EmpID: "8841"
  X-Automation-Origin: "AGENT_AUTOMATED_ACTION"

RESPONSE (200 OK — JSON SCHEMA BOUND):
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "emp_id": "8841",
  "full_name": "Sarah Jenkins",
  "contact_info": { "home_address": "123 High St, London, UK", "personal_phone": "+442079460921" },
  "remote_status": "REMOTE"
}
```

```
// [CONTRACT 2] POST /v1/employees/{emp_id}/leave — Submit Time-Off Request (FR-3.2 / FR-3.3)
REQUEST PAYLOAD (POST /v1/employees/8841/leave):
{
  "leave_type": "VACATION",
  "start_date": "2026-09-10",
  "end_date": "2026-09-12",
  "requested_hours": 24.0,
  "pre_check_signature": { "verified_remaining_hours": 96.0, "chronologic_order_valid": true }
}
```

4.2. ServiceImmediately ITSM REST API OpenAPI Contracts

```
// [CONTRACT 3] POST /v1/incidents – Auditable Support Ticket Creation (FR-4.1 / FR-4.3)
REQUEST PAYLOAD:
{
  "requestor_emp_id": "8841",
  "category": "Hardware",
  "priority": 3,
  "short_description": "Remote Home Office Monitor Dispatch - Remote Policy §3.2 Eligible",
  "dedupe_fingerprint": "SHA256(8841:Hardware:Remote Home Office Monitor Dispatch)" // Enforces 15m window
}
```

5. Technical Scalability Model: Concurrency, Connection Pooling & Burst Spikes

Models morning shift peak concurrency for 10,000 employees generating ~25,000 monthly queries with burst arrival spikes reaching **150 concurrent active sessions** (~25 req/sec):

System Tier / Resource	Autoscaling & Replicas	Connection Pooling & Multiplexing Strategy	Concurrency Cap & Overload Protection
Cloud Run ADK Orchestrator	min-instances = 2 max-instances = 20	Stateless HTTP/2 active multiplexing; concurrency = 80 reqs/instance.	Handles up to 1,600 concurrent turns before queuing; CPU limits 2 vCPU / 4 GiB per instance.
WorkWeek HCM REST Client	HTTP/2 Keep-Alive Client Pool	Shared connection pool capped at max 40 idle connections per replica (max 800 global).	Prevents saturating WorkWeek internal web server thread pool; 429 / 504 backoff retry (NFR-4.2).
ServiceImmediately ITSM Client	HTTP/2 Client Pool + Redis Fingerprint Hash	Shared pool (max 30 conns/instance); client checks Redis 15-minute deduplication hash prior to POST.	Eliminates duplicate ticket creation spikes during outage storms (FR-4.3).
Cloud SQL / PgBouncer DLQ Store	PgBouncer Connection Pooler	PgBouncer Transaction Pooling Mode with max_client_conn = 1000 and default_pool_size = 150.	Prevents PostgreSQL out-of-memory connection exhaustion; statement_timeout = 5000ms.

6. Comprehensive Tabular Error-Handling & Fallback Matrix (STAKEHOLDER GAP)

ID	Component Failure Scenario	Technical Resiliency & Compensating Action (NFR-4.2 / NFR-4.3)	Non-Technical User Notification Message (NFR-4.1)
ERR-01	WorkWeek API HTTP 429 / 504 Timeout	Exponential backoff retry (150ms → 400ms → 1000ms max 3 retries) with client-side token bucket.	"WorkWeek HCM is experiencing temporary delay. Retrying your request..." (Handled transparently; temporary service notice if retry exhausted).
ERR-02	Complete Backend Outage (HTTP 503)	Circuit breaker trips open. Internal stack traces masked.	"The HR Leave / Support Desk service is temporarily unreachable right now. Please try again in 5 minutes."
ERR-03	PTO Balance Exceeded / Validation Rejection (FR-3.3)	Client-side pre-invocation check aborts POST request before network execution.	"You currently have 12.0 hours of remaining Vacation PTO, which is insufficient for your request of 16.0 hours (2 days). Please adjust your requested dates."
ERR-04	Cross-System Chaining Partial Failure (UC-2.x)	Queues compensating record in Dead Letter Queue (DLQ) for automated IT dispatch.	"Your leave request (#LW-104) was submitted to WorkWeek. Opening your manager notification ticket in ServiceImmediately was delayed; our IT support team has been notified."
ERR-05	Policy Retrieval Confidence < 0.78 (FR-5.2)	Strict grounding checker aborts generation & sets POLICY_NOT_FOUND status.	"I could not find an explicit HR policy statement matching your query. Please open an HR inquiry ticket or consult your HR Business Partner."

ID	Component Failure Scenario	Technical Resiliency & Compensating Action (NFR-4.2 / NFR-4.3)	Non-Technical User Notification Message (NFR-4.1)
ERR-06	Prompt Injection / Jailbreak Detected (FR-1.3)	Sub-120ms Ingress Interceptor drops prompt & records SAFETY_BLOCKED audit event.	"Your request cannot be processed as it violates enterprise AI safety & HR domain guidelines."

7. Formal RBAC Permission Matrix & Identity Bridging Table

Enterprise Role Identity	WorkWeek Profile & PTO	WorkWeek Leave Requests	ServiceImmediately ITSM Tickets	Audit Logs & Vector Repo
Standard Employee (emp_id Scoped)	READ / UPDATE (Own only)	CREATE / VIEW (Own only)	CREATE / COMMENT (Own only)	READ Policy Repo. NO Audit access.
HR Manager / Approver	READ (Direct reports)	APPROVE / REJECT (Direct reports)	COMMENT / VIEW (Department)	READ Policy Repo.
IT Helpdesk Technician	NO ACCESS	NO ACCESS	UPDATE STATUS / RESOLVE	NO ACCESS to PII logs.
Agent Automation Identity	Scoped via Delegated Token	Scoped via Delegated Token	Requires Origin Header	WRITE ONLY (Audit Log Sink).
Audit Compliance Officer	NO ACCESS	NO ACCESS	NO ACCESS	READ ONLY (Masked BigQuery Logs).

8. Consolidated Technical & Operational Risk Register Table

ID	Risk Event & Description	Pre-Score	Concrete Architectural Counter-Measure & Mitigation	Post-Score
RSK-01	API Rate Limiting & Backend Throttling	HIGH (16)	Token bucket rate limiter + exponential backoff retry + Redis rate key.	LOW (3)
RSK-02	LLM Model Drift & Concept Drift	HIGH (12)	Automated CI/CD 4-tier golden evaluation gatekeeper blocking promotion.	LOW (2)
RSK-03	Policy Hallucination & Broken Citations	HIGH (15)	Hybrid RRF threshold ≥ 0.78 + mandatory deep-link verification.	LOW (2)
RSK-04	Prompt Injection / State Override	CRITICAL (20)	Sub-120ms dynamic interceptors + HMAC signed Delegated Token.	LOW (3)
RSK-05	Cross-System Chaining Partial Failure	MEDIUM (9)	Automated Dead Letter Queue (DLQ) compensating rollback.	LOW (2)
RSK-06	SPII Data Leakage into Log Sinks	CRITICAL (20)	Local regex/NER masking replacing SPII with [REDACTED_SPII].	LOW (2)

9. Terraform IaC Code Snippets & CI/CD Pipeline YAML Gatekeeper (STAKEHOLDER GAP)

9.1. Complete Terraform IaC Configuration Snippet (main.tf & iam.tf)

```
# [main.tf] VPC Service Controls Perimeter & GKE Autopilot / Cloud Run Service
resource "google_cloud_run_v2_service" "hr_agent_orchestrator" {
  name      = "hr-agent-orchestrator-mvp1"
  location  = "europe-west1"
  ingress   = "INGRESS_TRAFFIC_INTERNAL_ONLY" # Enforces Private VPC Service Controls Ingress

  template {
    service_account = google_service_account.hr_agent_sa.email
    scaling {
      min_instance_count = 2
      max_instance_count = 20
    }
    containers {
      image = "eu.gcr.io/altostrat-hr-ai/hr-agent-orchestrator:v1.3"
      resources { limits = { cpu = "2", memory = "4Gi" } }
      env { name = "WORKWEEK_API_BASE_URL"; value = "https://workweek-internal.altostrat.corp/v1" }
      env { name = "ITSM_API_BASE_URL"; value = "https://itsm-internal.altostrat.corp/v1" }
      env { name = "REDIS_HOST"; value = google_redis_instance.session_store.host }
    }
  }
}

# [iam.tf] Workload Identity Federation Role Bindings
resource "google_service_account_iam_member" "workload_identity_binding" {
  service_account_id = google_service_account.hr_agent_sa.name
  role                = "roles/iam.workloadIdentityUser"
  member              = "serviceAccount:altostrat-idp.svc.id.goog[hr-namespace/hr-agent-service]"
}
```

9.2. Automated CI/CD Pipeline Configuration YAML (cloudbuild.yaml with ADK Eval Gatekeeper)

```
steps:
  # STEP 1: Lint OpenAPI Contracts & Tool Definitions
  - name: 'stoplight/spectral'
    args: ['lint', 'contracts/workweek_openapi.yaml', 'contracts/itsm_openapi.yaml']

  # STEP 2: Execute ADK Golden Evaluation Suite on Sandbox Mock
  - name: 'gcr.io/altostrat-hr-ai/google-agents-cli:latest'
    args:
      - 'eval'
      - 'run'
      - '--config=eval_config.json'
      - '--evalset=evalsets/4_tier_golden_evalset.json'
      - '--output=eval_results.json'

  # STEP 3: Automated Quality Gatekeeper Assertion Script
  - name: 'python:3.13-slim'
    script: |
      import json, sys
      res = json.load(open('eval_results.json'))
      print(f"Policy Q&A Accuracy: {res['metrics']['policy_accuracy']}% (Target >= 95%)")
      print(f"Hallucination Rate: {res['metrics']['hallucination_rate']}% (Target == 0%)")
      print(f"Safety Block Rate: {res['metrics']['safety_block_rate']}% (Target == 100%)")
      print(f"Cross-System Pass: {res['metrics']['cross_system_pass_rate']}% (Target == 100%)")

      if res['metrics']['policy_accuracy'] < 95.0 or res['metrics']['hallucination_rate'] > 0.0:
        print("CRITICAL: ADK Evaluation Gatekeeper FAILED! Promotion blocked.")
        sys.exit(1)
      print("ADK Evaluation Gatekeeper PASSED. Proceeding to Staging Rollout.")
```


10. FinOps Cost Estimation Model & Agentic Development Schedule (STAKEHOLDER GAP)

10.1. Itemized FinOps Cost & Scaling Model Table (10,000 Employees | ~25,000 Queries/Mo)

Cost Component	GCP Enterprise Tier Pricing	Estimated Monthly Usage	Monthly Cost Estimate
Gemini 1.5 Pro / Flash (Core LLM)	\$1.25 / M input • \$5.00 / M output tokens	30M Input • 6.25M Output Tokens	\$37.50 (Input) + \$31.25 (Output)
Vertex AI Vector Search Index	Standard Shard Hosting	1 Active Index Shard	\$145.00
Cloud Run / GKE Autopilot Compute	4 Replicas (2 CPU, 4 GB RAM)	99.9% Uptime SLA	\$180.00
BigQuery & Cloud Logging (Audit Sink)	\$0.01 / GB Ingested	50 GB Audit & Trace Logs	\$0.50
TOTAL MONTHLY OPERATING COST (~25,000 queries/mo)			~\$394.25 / mo (~\$0.016 / query)
BUSINESS ROI: 40% TICKET DEFLECTION (10,000 Tickets/Mo Deflected @ ~\$18/Ticket)			\$180,000 / mo Gross Savings

10.2. Agentic Software Engineering Delivery Schedule (4-Sprint Roadmap)

Sprint & Dates	Focus Area & Core Deliverables	Specific Engineering Tasks & ADK Subagent Scaffolding	Verification Gate & Quality Criteria
Sprint 1 (Weeks 1-2)	Traceably Bounded Sandbox & Guardrails	Scaffold supervisor_agent, dynamic interceptors (FR-1.3, FR-1.4), Terraform VPC perimeter, BigQuery WORM audit sink (FR-1.2), & Tier 1/4 evalsets.	100% Ingress Safety Block on prompt injections; 0ms PII leak.
Sprint 2 (Weeks 3-4)	WorkWeek & ITSM Subagents	Implement HMAC Delegated Auth Scoping (FR-3.1), pre-check balance validators (FR-3.3), ITSM transition graph check (FR-4.3), & Redis post-turn PII eviction (FR-3.4).	100% Transaction Correctness on PTO balance checks.
Sprint 3 (Weeks 5-6)	Hybrid Policy RAG & Citations	Ingest & chunk HR PDF policies preserving section anchor URLs (#sec-3.2), wire Hybrid RAG + RRF, & configure Cloud Storage Object Change Webhook Watchdog (FR-5.5).	Tier 1 Policy Q&A Accuracy ≥ 95% on golden evalset; 0% hallucinations.
Sprint 4 (Weeks 7-8)	Cross-System Chaining & Staging UAT	Orchestrate multi-step chains (UC-2.1 - UC-2.3), wire DLQ compensating rollback (NFR-4.3), execute CI/CD gatekeeper suite, & complete enterprise UAT.	100% Pass on all UC-2.x cross-system chains; sign-off for Staging rollout.

11. Open Questions, Known Unknowns & Post-MVP Hypotheses (STAKEHOLDER GAP)

To ensure complete operational transparency and prevent presenting the MVP design as static or unvalidated, this section documents tracked architectural unknowns, operational risks under investigation, and empirical post-MVP validation hypotheses.

11.1. Tracked Open Questions & Known Unknowns Table

ID	Tracked Open Question / Known Unknown	Architectural Implication & Investigation Scope	Assigned Owner	Resolution Gate
OQ-01	Policy Sync Webhook Quotas (FR-5.5) Can Cloud Storage Object Change Webhooks trigger real-time Vertex AI Vector Search re-indexing (<15m SLA) during bulk policy refactors?	Investigate Vertex AI streaming index update quotas under simultaneous multi-document commits to prevent index queue backpressure.	AI Platform Lead	Sprint 1 Gate
OQ-02	Granular Medical SPII Routing (UC-2.2) Should short-term medical leave notes be redacted completely from ServiceImmediately ticket descriptions, or routed to a restricted HR Medical queue?	Determine whether Egress Interceptor (FR-1.4) should mask diagnosis text to [REDACTED_MEDICAL_SPII] before ITSM ticket POST.	SecOps & Legal	Sprint 2 Gate
OQ-03	WorkWeek On-Prem Thread Pool Capacity What is the exact HTTP connection limit of WorkWeek's legacy API gateway during 9:00 AM shift start bursts (150 concurrent sessions)?	Conduct load simulation against WorkWeek staging endpoint to fine-tune PgBouncer & HTTP/2 keep-alive connection pool sizes.	HCM Integrations Lead	Sprint 2 Gate
OQ-04	Multi-Region Anycast Subagent Roundtrips Will Anycast Global LB routing APJ employees to European active-active GKE Autopilot clusters exceed the <10s TTFT SLA?	Benchmark open telemetry distributed trace spans across GKE Europe and APJ regions during cross-system chaining (UC-2.1).	GCP Infra Architect	Sprint 3 Gate

11.2. Post-MVP Empirical Hypotheses & Validation Metrics

EMPIRICAL POST-MVP LEARNING LOOP (AGENTIC DEVELOPMENT)

Each hypothesis below is instrumented via BigQuery WORM audit telemetry (audit_event_traces) and evaluated automatically during bi-weekly Enterprise Architecture Review Board sprints.

Hypothesis ID & Title	Core Post-MVP Hypothesis Statement	Telemetry Instrument & Success Criterion
HYP-01 Deflection Elasticity	We hypothesize that enforcing client-side PTO balance pre-checking (FR-3.3) and 15-minute duplicate ticket hash scanning (FR-4.3) will eliminate >90% of erroneous leave/ticket submissions, driving overall Tier 1 helpdesk deflection from 25% to ≥ 40% within 180 days .	Compare monthly SUBMITTED vs PRE_CHECK_FAILED audit events against total helpdesk ticket volume (≥ 10,000 deflected tickets/mo).
HYP-02 Zero-Cache Latency Trade-Off	We hypothesize that fetching Employee Profile and PTO balances live on every query turn (FR-3.4) adds < 180ms overhead to overall turn latency while eliminating 100% of stale leave approval edge cases and meeting GDPR Article 17 zero-retention compliance.	OpenTelemetry P95 span duration on workweek_subagent live API calls vs. total dialogue turn latency (< 10.0s TTFT target).

Hypothesis ID & Title	Core Post-MVP Hypothesis Statement	Telemetry Instrument & Success Criterion
HYP-03 Hybrid RRF Grounding Superiority	We hypothesize that combining BM25 keyword retrieval with Vertex AI text-embedding-004 semantic search via Reciprocal Rank Fusion (RRF) increases exact HR Policy Q&A exact grounding accuracy (FR-5.2) from 88% to ≥ 95.5% , reducing policy hallucinations to 0.0%.	Automated LLM-as-a-Judge (eval-adk-skill) golden evaluation score across Tier 1 policy test suite (target 0% hallucination).